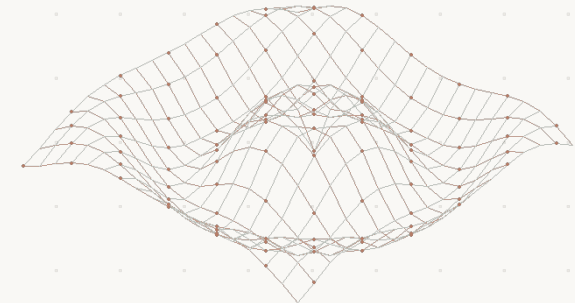


Callout

Real-time voice comms translation for competitive gaming squads.

Each player speaks naturally in their native language; teammates hear the callout re-synthesized in their own language in under one second.



Core Stack: Python · FastAPI · WebSockets · AssemblyAI Streaming · Groq LPU · Web Audio API

The Failure Modes of Conventional Translation in Esports

Why standard machine translation and text-based tooling fall short during live gameplay.



01 / Temporal Latency

A two-second lag is equivalent to silence

In modern competitive shooters, round-deciding engagements occur in fractions of a second. Traditional translation pipelines introduce latency that renders critical callouts obsolete before they arrive.

02 / Tactical Vernacular

Jargon and dialect resist general NLP

Esports communication relies on dense shorthand—terms like 'one-shot', 'diff', 'clutch', or 'flank'. Literal translation tools misinterpret gaming idiom as literal prose, stripping out tactical value.

03 / Multilingual Code-Switching

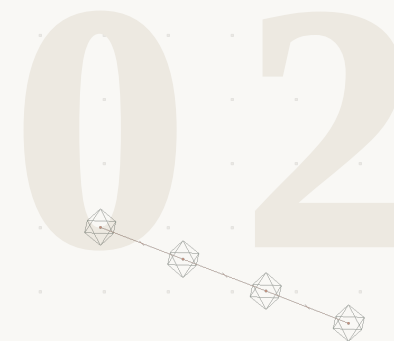
Natural speech fluidly mixes vocabularies

Players frequently alternate between native languages and English loanwords mid-sentence (e.g. Hinglish). Standard models fail when sentences lack rigid grammatical syntax.

04 / Cognitive & Physical Demands

Hands and eyes must remain on the match

Gamers cannot take their fingers off movement keys to read subtitles or type in team chat. Voice is the sole viable channel for tactical coordination.



02 — METHODOLOGY

Acoustic Translation in Four Coordinated Stages

How voice moves from a player's microphone to teammates' headsets in under a second.

Phase 01

Client-Side Audio Sampling

The browser captures microphone input via the Web Audio API, downsamples the stream to 16 kHz mono PCM, and transmits chunked binary packets over an authenticated WebSocket.

Phase 02

Streaming Speech Recognition

AssemblyAI processes live audio chunks asynchronously. The model dynamically detects the spoken language and isolates discrete conversational turns via speech-boundary markers.

Phase 03

Domain-Aware Translation

Groq LPUs execute few-shot tactical prompt formatting in sub-200ms. Conversational text is rewritten into concise, punchy callouts under ten words in each teammate's chosen language.

Phase 04

Individualized Audio Delivery

Callouts are broadcast back to client browsers. Each listener hears natural speech via system speech synthesis or streamed high-fidelity ElevenLabs audio.

End-to-End Latency Budget & Distributed Architecture

A decoupled, asynchronous architecture engineered for predictable performance under load.

— Edge Client (Browser)

Vanilla Web Audio API · AudioContext downsampler · Full duplex WebSocket connection

— Orchestration Hub (FastAPI)

Room session coordination · Concurrency control · In-memory translation caching

— Speech Processing (AssemblyAI)

Universal-3.5-pro streaming engine · Real-time turn demarcation · Keyterm bias

— Inference Engine (Groq)

Qwen 3.8-27B hosted on Groq LPUs · Multi-language batch generation · Sub-200ms TTFT

— Speech Synthesis (Hybrid)

Native Web Speech API with Latin romanization · Optional streamed ElevenLabs Flash

LATENCY BUDGET BREAKDOWN

Speech Boundary Detection : 300 – 450 ms

Streaming turn finalization via AssemblyAI

LLM Translation Inference : 120 – 180 ms

Fast few-shot generation on Groq LPU hardware

Network Roundtrip (WS) : 40 – 80 ms

Full-duplex WebSocket broadcast to teammates

Audio Synthesis & Playback : 80 – 150 ms

Local browser synthesis or stream buffering

Total Measured Latency : $\approx$ 750 – 950 ms

Consistently sub-second across tests



04 — DOMAIN ADAPTATION

Linguistic Precision for Competitive Environments

Specialized prompt engineering, colloquial priming, and automated phonetic transliteration.

01 / Concise Tactical Syntax

Enforcing ten-word limits with zero filler

The translation model is strictly instructed to strip conversational hesitations ('uh', 'you know') and distill raw dialogue into ten words or fewer. Messages retain urgent, directive phrasing natural to high-pressure gameplay.

02 / Esports Terminology Glossary

Preserving recognized tactical vocabulary

The system maintains contextual priming for key terms such as 'flank', 'one-shot', 'diff', 'clutch', 'rotate', and specific map callout names. Terms remain in their recognized tactical form across all supported languages.

03 / Phonetic Transliteration

Bridging non-Latin scripts to native system voices

Many desktop operating systems lack natural voice synthesizers for scripts like Devanagari or Arabic. The server automatically romanizes these outputs into intuitive phonetic Latin text before transmission, ensuring crisp playback on standard system speech engines.

Tactical Voice Agent: Hands-Free Comms Retrieval

Private audio co-pilot providing contextual recall without squad interruption.



COMMAND 01 / REPLAY

“Agent, repeat”

Instantly replays the most recent teammate callout translated into the listener's language.

Resolves a critical problem in competitive shooters where in-game gunfire, explosions, or footsteps drown out voice communications during pivotal moments.

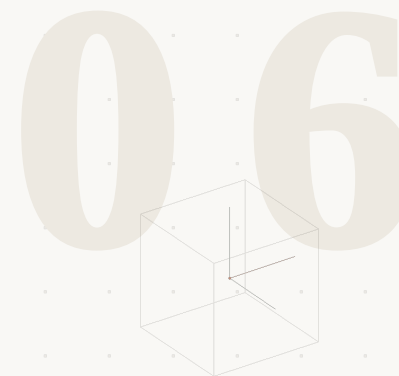
COMMAND 02 / SYNTHESIS

“Agent, summary”

Condenses the previous eight callouts across the room into a single 25-word situation brief.

Allows players rotating, clutching, or defusing to re-orient themselves to the tactical state of the round without requiring teammates to stop and explain.

Architectural principle: Agent responses are delivered privately to the inquiring player and never broadcast to teammates.



06 — SPECIFICATIONS

Technical Stack & Operational Benchmarks

Core framework selections and measured performance parameters.

— Server & Transport

Python 3.12 · FastAPI asynchronous application framework · Uvicorn ASGI server · Native WebSockets

— Speech Ingestion

AssemblyAI Streaming API v3 · Persistent PCM client · Real-time turn boundary detection

— Model Inference

Groq Cloud LPU platform · Qwen 3.8-27B · In-memory translation caching

— Audio Synthesis

Client-side Web Speech API with Latin romanization · Streamed ElevenLabs Flash fallback

— User Interface

Lightweight HTML5 and CSS3 · Zero external UI frameworks · Three.js interactive 3D hero canvas

OPERATIONAL BENCHMARKS

< 1.0 second

End-to-End Latency — Speech input to translated audio delivery

10+ Languages

Broad Multilingual Coverage — English, Hindi, Spanish, French, German, Japanese, Arabic, etc.

6 Players

Room Capacity — Concurrent per-player streaming and audio routing

Zero Overhead

Browser Native — Runs in any modern browser without software installation



Live Demonstration Protocol & Strategic Expansion

Demonstrated real-world efficacy and the path toward native gaming ecosystem integration.

LIVE DEMONSTRATION SCRIPT

01 / Session Initialization

Host creates a room code on primary desktop; peer joins via mobile browser.

02 / Unidirectional Translation

Desktop user speaks colloquial Hindi callout; mobile device plays back English in sub-second time.

03 / Bidirectional Response

Mobile user replies in English; desktop audio outputs natural translated Hindi.

04 / Tactical Voice Agent

Demonstrating 'Agent, repeat' and 'Agent, summary' voice wake-word triggers.

STRATEGIC EXPANSION

— Transparent In-Game Overlay

Packaging as an Overwolf / Electron desktop client to display unobtrusive visual callouts over active full-screen titles.

— Discord Bot Integration

Deploying as a headless voice channel participant to translate squad voice directly within existing Discord workflows.

— Acoustic Noise Suppression

Incorporating lightweight client-side noise models (e.g. RNNoise) to filter out mechanical keystrokes and system fan noise.

Thank You & Open Tactical Q&A

Full-duplex multilingual translation architecture ready for live squad deployment.

KEY ARCHITECTURAL ACHIEVEMENTS

01 / Sub-Second End-to-End Latency

Streaming 16 kHz PCM over WebSockets with AssemblyAI Universal-3.5 and Groq LPUs achieves sub-850ms turnaround, preserving competitive reaction times.

02 / Esports Shorthand & Code-Switching

Domain-tuned prompting decodes dense shorthand (flank, one-shot, rotate, diff) and natural multilingual code-switching (e.g. Hinglish) without failure.

03 / Hands-Free Voice Agent Intelligentsia

Autonomous private trigger phrases ("*Agent, repeat*", "*Agent, summary*") synthesize immediate audio situational debriefs without team chatter.

DEPLOYMENT & DISCUSSION

— Open-Source Codebase & Docs

FastAPI hub, 3D WebGL slide deck, tactical voice HUD, and complete setup blueprints available on GitHub:

github.com/daarcbro/callout

— Zero-Install Squad Testing Terminal

Instant room creation with shareable squad codes, continuous mic streaming, and Web Speech / ElevenLabs dual synthesis.

— Open Floor for Questions & Demo

Thank you for reviewing the Callout project. We welcome your technical questions, feedback, and live demonstration requests.

